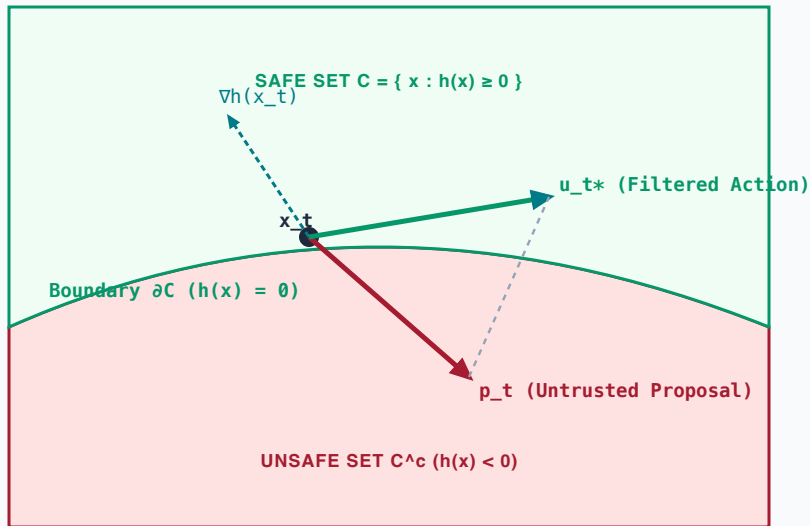


MINIMAL-INTERVENTION CONTROL BARRIER FUNCTION (CBF) PROJECTION

State space forward invariance $\dot{h}(x, u) \geq -\alpha(h(x))$ and closed-form quadratic program action projection

1. STATE SPACE GEOMETRY (FORWARD INVARIANCE OF C)



FORWARD INVARIANCE GUARANTEE:

- By enforcing Nagumo's condition via CBF ($L_f h + L_g h u \geq -\alpha(h)$), the system guarantees that state x_t never breaches the boundary ∂C .
- Preserves maximum task intent while rigidly enforcing safety.

2. ACTION SPACE QP PROJECTION SOLVED IN 65 μ s ON MCU

MINIMAL-INTERVENTION QUADRATIC PROGRAM (QP)

$u_{t*} = \operatorname{argmin}_{\{u\}} (1/2) ||u - p_t||_2^2$
subject to: $a^T u \leq b$ (Affine Half-Space)
where $a = -L_g h(x)$ and $b = L_f h(x) + \alpha(h(x))$

CLOSED-FORM ANALYTICAL PROJECTION (ZERO ITERATIONS)

if ($a^T p_t \leq b$) return p_t ; // Proposal is Safe
else $u* = p_t - [(a^T p_t - b) / ||a||_2^2] \cdot a$

- Evaluates 1 scalar dot-product and 1 vector subtraction.
- Execution latency: **65 microseconds** on Cortex-M4.

SYSTEMS METROLOGY ADVANTAGE

- **Deterministic Bare-Metal:** Zero matrix inversion, zero numerical convergence loops, zero solve latency.
- **Continuous 1000 Hz Invariant:** The MCU reflex operates with guaranteed microsecond latency.